

Game-Based Digital Math Intervention's Effect on Math Anxiety and Its Variation Across Classrooms

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Digital math learning interventions, especially game-based interventions, have the potential to reduce math anxiety in students; however, evidence of their impact is mixed. We examined the effects of a game-based and a non-game-based digital math learning intervention on 677 seventh-grade students' math anxiety and how these effects vary across classrooms. Although math anxiety did not decrease as a result of the interventions, our analysis demonstrated large variation of the intervention type effect between classrooms, indicating some classroom characteristics might significantly impact the way interventions influenced math anxiety. Moreover, a small but significant cross-level interaction between intervention type and classroom average math anxiety was detected, where students in classrooms with higher levels of math anxiety demonstrated a larger decrease in math anxiety after solving problems in a non-gamed based vs. gamed-based intervention. Potential explanations and implications of findings are discussed.

Data, analysis code, supplementary material: <https://osf.io/89dnm/>

Keywords: math anxiety, math learning, game-based learning, digital learning, hierarchical linear models

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Introduction

A primary goal of math educators has been to improve student experiences with mathematics and reduce math anxiety — defined as apprehension, nervousness, or fear related to math (Ashcraft, 2002). Math anxiety is not simply a dislike of math; it often encompasses strong emotional and physiological reactions to math (Ramirez et al., 2018), with fMRI studies finding that math-anxious individuals show similar neurological reaction patterns to math that one would expect to see when anticipating bodily harm or visceral pain (Lyons & Beilock, 2012). With such strong reactions, it is not surprising that math anxiety has been found to detrimentally impact arithmetic performance (Van Mier et al., 2019), basic number processing (Maloney et al., 2011; Skagerlund et al., 2019), and predict poor math achievement in general (Pantoja et al., 2020). Studies have demonstrated that higher math anxiety is also related to avoidance behaviors, such as taking fewer math-related courses and pursuing fewer science, technology, engineering, and math-related occupations (Choe et al., 2019). As it is estimated that nearly 33% of 15-year-old students worldwide experience regular feelings of math anxiety (OECD, 2013), math anxiety is a considerable challenge for many students.

Several theoretical frameworks offer insight into how math anxiety develops and suggest various mechanisms to explain the math anxiety and performance connection. For example, the Reduced Competency Account (Maloney, 2016) argues that students with higher math anxiety show poorer performance because they lack mathematical skills, which leads to them repeatedly experiencing failures in math-related tasks, leading to negative attitudes and anxiety towards math. This account further suggests that poor math abilities could lead to avoidance of math-related activities which would worsen performance and math anxiety even more (Ramirez et al., 2018). In contrast, the Disruption Account (Ashcraft & Kirk, 2001) argues that intrusive worries caused by anxiety take up critical working memory resources needed to solve math problems resulting in lower math performance. Thus, according to the Disruption Account, it is the anxiety itself that prevents students from performing well, rather than a lack of knowledge or skills in mathematics. Although these two accounts point to different mechanisms in their explanations of the relationship between math anxiety and performance, evidence has been found for both the Reduced Competency (Ma & Xu, 2004; Maloney et al., 2011) and Disruption Account (Ashcraft & Kirk, 2001), signaling that both mechanisms may be at play (Carey et al., 2016).

In recent years, a new framework of math anxiety was proposed that moved away from explaining the math anxiety-performance link to explaining why some students develop math anxiety and others do not. According to the Interpretation Account (Ramirez et al., 2018), it is not just poor math skills that may lead a

student to develop math anxiety, but their individual interpretation of disfluent math experiences (e.g., struggle, failure) that can create math anxiety. For example, if two students get a problem wrong on a math test, one might appraise the failure as reflecting their inability to do math, while the other interprets the failure as simply the problem being hard. Evidence for the interpretation account includes research finding that negative appraisals of math shape students' attitudes toward math (Hunt & Maloney, 2022) and providing negative feedback (opposed to positive or neutral feedback) can cause students to negatively interpret their math performance, leading to worsening cognitive, emotional, and physiological states (Mammarella et al., 2023).

Collectively, these three accounts signal important opportunities for intervention and lead to an important question: How can teachers and digital interventions best improve math skills, reduce intrusive worries or reduce unnecessary cognitive load, and/or help students to interpret struggle and failure more positively?

Game-based digital learning platforms as a means to decrease math anxiety

Game-based digital learning – the use of games, or structured play with goals, rules, and challenges that are employed to diverse and amuse individuals (Cheng et al., 2015; Roedavan et al., 2021) – have increased dramatically in recent years. Although there is no agreed-upon taxonomy for what exactly constitutes a gamified element or technique (Dichev & Dicheva, 2017; Toda, Klock, et al., 2019), Garris et al. (2002) suggest that game-based learning platforms include some level of fantasy context with imaginary sensory design and given clear goals, rules, and feedback systems on their optimal level of challenge with some ability to control it and some level of mystery. Such game mechanics and elements aim to provide entertaining content while expanding students' knowledge.

Based on the three aforementioned theoretical frameworks of math anxiety (Reduced Competency Account, Disruption Account, and Interpretation Account), we see several potential opportunities where game-based platforms could support math anxiety reduction in students. First, game-based digital learning platforms may help decrease math anxiety by improving students' math skills and personal experiences with math. Having better performance as a result of playing, students will gain more positive experience with math which might improve their math attitude and decrease math anxiety. Indeed, research has shown that game-based digital learning platforms can improve math skills (Hussein et al., 2022; Tokac et al., 2019), and tutoring and improving math skills can reduce math anxiety for students (e.g., Supekar et al., 2015), which aligns with Reduced Competency account of math anxiety development.

Moreover, game-based digital learning platforms may additionally help students to become and remain more engaged with the content, disrupting cycles of avoidance and poor math skills. Indeed, many researchers refer to Self-Determination Theory (Ryan & Deci, 2000) as the theoretical basis for game-based platform usage for learning; the theory posits that competency, autonomy, and relatedness are core psychological needs that improve intrinsic motivation when met (Burgers et al., 2015; Nacke & Deterding, 2017; Songer & Miyata, 2014). A Reduced Competency account of math anxiety suggests that if game-based digital learning platforms increase motivation and engagement in students (Toda, Palomino, et al., 2019), students may exhibit less avoidance behavior and overcome anxious responses to math (Chang & Beilock, 2016), thus develop greater skills in math which, in turn, would reduce their anxious attitude towards math.

A third aspect of game-based digital learning platforms that could theoretically reduce math anxiety is through the change of context. They provide math activities through a new context that could be less triggering compared to a normal classroom setting that students typically associate with nervousness and anxiety. In fact, some of the validated math anxiety scales ask about typical classroom environments that students find stressful, including watching an instructor solve an algebra problem on the board or starting a new chapter in a math book (Hopko et al., 2003). Game-based learning platforms can change this context. For instance, one of the distinct techniques of game-based learning that educators frequently use is narrative/storytelling that describes fictional worlds. For students who play *DimensionM*, they are not simply completing a math assignment but navigating new planets as space explorers (Offenholley, 2012). The change of context through game-based interventions can not only engage learners by creating additional meaning to their experience (Kapp, 2012) but offer a novel context that students do not associate with math anxiety.

A fourth way that game-based learning platforms can reduce math anxiety is through the creation of a safer environment to explore math. Most games normalize failures and multiple attempts in achieving goals by providing an opportunity to restart and replay the game as many times as the individual wants. In a learning environment, the replay option might support students in testing various hypotheses about the material, exploring solving approaches, and testing which approaches could be more effective than others (Kapp, 2012). The Interpretation Account of math anxiety argues that this shift in struggle interpretation may be especially effective in helping students to alleviate their math anxiety.

Finally, various game mechanics could decrease students' math anxiety by helping them build self-confidence. Games could utilize progress bars that show how many goals a gamer has already reached, and rewards – badges, points, and other virtual items – that

represent students' achievement in specific areas. These elements could emphasize each gamer's accomplishment – even smaller ones – so even lower-achieving students have some positive reinforcement, can reinterpret their work, and focus on their positive experiences more. This might raise students' confidence, which has been associated with lower math anxiety (Desender & Sasanguie, 2022).

Effectiveness of game-based interventions on math anxiety

Although there are theoretical justifications for how game-based learning platforms may reduce math anxiety, previous research is mixed, having found both positive effects of such platforms on reducing math anxiety (Alanazi, 2020; Maissey et al., 2022; Ng et al., 2022), as well as no significant impacts (Hung et al., 2014; Jansen et al., 2013). In a recent meta-analysis, Dondio et al. (2023) analyzed the results of 22 experiments with game-based digital interventions and demonstrated that digital games had a marginally significant small effect on reducing math anxiety. Specifically, only five interventions significantly reduced math anxiety and others did not demonstrate significant effects. Importantly, none of the studies demonstrated that playing game-based learning platforms worsened math anxiety. Moreover, the authors noted that the game-based intervention effect on math anxiety was moderated by how long the game was played: specifically, the longer learners played the greater effect was detected, which is in accordance with other studies that found a similar effect on math performance (Lavasani & Khandan, 2011). However, most studies examining the effect of game-based math learning interventions on math anxiety do not analyze how that effect varies across classrooms and what classroom-level factors can moderate such an effect.

Importance of classroom factors

It is likely that specific characteristics of the classrooms can affect the math anxiety of students in those classrooms and the way game-based math learning platforms influence attitudes towards math. For example, OECD (2013) highlighted the role of classroom-level performance in individual math attitudes, specifically through social comparisons: across the majority of countries participating in the OECD's Programme for International Student Assessment, including the US, the better the schoolmates' performance, the lower the levels of students' self-concept and the more likely they were to develop math anxiety. In addition to peer comparison, climate can also play a role. For instance, a negative peer climate can decrease autonomous motivation in math (Reindl et al., 2015), while perceived student cohesiveness, cooperation and task orientation in the classroom were negatively associated with anxiety about mathematics learning (Taylor & Fraser, 2013), as well as

positively associated with students' self-esteem in math (Chionh & Fraser, 2009).

Kim et al. (2023) further demonstrated that students tend to become more similar in their math attitudes to those of their friends. Researchers suggested that students can transmit math anxiety to their peers through emotional contagion and co-rumination or modeling and mimicking behaviors related to math, which is in accordance with the Interpretation Account of math anxiety development. Similarly to peer contagion related to math attitudes (Kim et al., 2023), students can be impacted by their peers' emotions towards game-based math learning interventions. For example, if students in a classroom were assigned to two types of conditions – a game-based condition and a more classical math learning condition – and the majority of students in that classroom demonstrate high levels of math anxiety, students can transfer less worry and concern against game-based learning to each other. Then, students assigned to game-based learning intervention might have more positive attitudes towards their intervention and intervention content – math – in general. However, to our knowledge, researchers have not yet studied such a moderation effect of classroom-level factors on the association between game-based interventions and math anxiety.

The Current Study

In the current study, we investigated how game-based and non-game-based digital math learning interventions affect math anxiety, and if the characteristics of their classrooms impacted that effect. Specifically, we compared the effect of two digital math learning platforms on middle schoolers' math anxiety: the game-based math learning platform called From Here To There (FH2T) and a limited version of a non-game-based math learning tool called ASSISTments. Both platforms have been shown to positively impact math performance in students (Decker-Woodrow et al., 2023; Roschelle et al., 2016), which, according to the Reduced Competency account, can lead to less math anxiety. The impact of these interventions on math anxiety, however, has yet to be examined.

Thus, we asked two research questions. First, how does participating in game-based and non-game-based math learning interventions affect students' math anxiety? And second, do classroom characteristics impact the effect of the intervention type (game-based or not game-based) on post-intervention math anxiety?

For the first research question, we hypothesized that math anxiety (MA) would decrease after both interventions (H1a). Moreover, we hypothesized that math anxiety would decrease more for students in the "From Here to There" (FH2T) condition than the ASSISTments condition (H1b). We speculated that the game-based nature of FH2T provides additional features and new context that could further reduce math anxiety. For the second research question, we hypothesized that

the effect of the intervention type on post-intervention math anxiety would largely vary across classrooms (H2a). In addition, we hypothesized that classroom average pre-intervention math anxiety would moderate the association between the intervention type and individual post-intervention anxiety, such that students who were in a game-based condition and who were in higher math-anxious classrooms would demonstrate a steeper decrease in math anxiety compared to students in a non-game-based condition who were in a less math anxious classrooms (H2b).

Hypotheses H1a and H1b were preregistered on the OSF platform (<https://osf.io/89dnm/>), all other analyses should be considered exploratory. In addition, in the preregistration, we hypothesized that individual pre-intervention math anxiety would moderate the association between the intervention type and individual post-intervention anxiety, such that students with higher pre-intervention anxiety would demonstrate a steeper decrease in math anxiety after game-based intervention than students with lower pre-intervention anxiety. Our data did not demonstrate this moderation effect to be significant. Thus to keep our paper more concise, full analyses with associated tables are reported in supplemental materials on the OSF platform.

Methods

Dataset and Participants

Our data come from a larger randomized controlled trial study (Ottmar et al., 2023). This larger study compared the effects on math learning of three digital math learning interventions versus an active control condition: FH2T, DragonBox, and problem sets with immediate hints and feedback in ASSISTments. FH2T and DragonBox are both game-based platforms, whereas ASSISTments involves more traditional textbook math problems (ASSISTments is described in detail below). In summary, the impact was found for the two game-based platforms compared to the active control. The full impact results of that larger efficacy study are described in detail in previous work (Decker-Woodrow et al., 2023).

The larger study recruited 4,092 7th-grade students (12-13 years old) from 11 middle schools in a large suburban district in the Southeastern United States in the 2020-2021 academic year. Parents were notified of the study and had the option to withdraw their children from the study. As this study occurred during the onset of the COVID-19 pandemic, students had the option to attend school in person or virtually. Only children who attended school in person were included in the sample for the current paper.

Students were randomly assigned to one of four conditions (FH2T, DragonBox, ASSISTments, active control) for the full school year. Initially, we planned to analyze three study conditions (FH2T, ASSISTments, and Dragon Box), however, given the disruption of the COVID-19 pandemic and changes in remote learning,

there were major technical issues experienced by the majority of Dragon Box users when trying to use the program. For example, as DB is a commercially available downloadable app that is not available on the cloud, many students were unable to install the program on a personal device at home for homework or had access to their created profiles when switching from virtual to in-school learning. These challenges resulted in significantly higher attrition in the DB condition. Upon closer analysis, the participants who were assigned to Dragon Box conditions but dropped off from the study had marginally significantly higher levels of pre-intervention math anxiety than participants from the ASSISTments condition ($p = .05$). Thus, the decision was made to only analyze data from students who were assigned to the FH2T and ASSISTments.

Due to the disruption and movement between virtual and in-school learning, we also only included those students who attended in-person classes during the whole intervention. We then deleted observations with at least one missing measurement of pre- or post-intervention math anxiety or pre-intervention math performance. In the final sample of 677 students from 98 classrooms, 54% were male and 46% female. The race/ethnicity was 67% White, 19% Hispanic, 6% Asian, and 4% reported two or more races. The smallest classroom had 1 student included in our sample, the largest had 25 students.

Materials

Data from two technology platforms were used in this study: From Here to There! (FH2T) and ASSISTments (described below).

From Here to There!

In our study, we used a game-based digital learning platform, From Here to There! (FH2T) (Ottmar et al., 2015) to examine whether math games could reduce math anxiety. As a game-based platform, FH2T uses gamified visual graphic design and proposes students to solve math problems as part of a larger story, where they go up through a math “tree” addressing 14 math “worlds” (see Figure A1) that each focus on a different math concept with increased levels of difficulty. Instead of a traditional math problem, each problem in FH2T presents as a puzzle with a start state and a goal state, where students can manipulate algebraic expressions with gestures to transform it from the start to the goal state. For example, students could be asked to convert “ $14 \times 2 \times 7y \times 1 \times 4$ ” to the goal state “ $28y \times 1 \times 8$ ” by dragging numbers with the computer mouse to appropriate places (see Figure A2). The design of the game came from theories of perceptual learning (Goldstone et al., 2010) and embodied cognition (Abrahamson et al., 2020), as well as research in math education and cognition.

With the feedback and scaffolding provided in the game, FH2T was intended to provide a safe environment for

making mistakes and learning from them. The platform provides soft feedback on students’ problem-solving effectiveness; specifically, it gives all students clovers as rewards, and the number of clovers symbolizes the level of student’s solving efficiency (see Figure A3). Those students, whose performance on a specific problem was not the most efficient, are able to replay the problem as many times as they want and each time get an updated number of clovers without any damage to their game progression. Moreover, the gameplay in FH2T allows students to attempt an invalid mathematical action, but does not let students commit mistakes – when users try to make mathematically incorrect actions, the system provides a signal that this action could not be performed by shaking the terms with which the student interacted and resetting back to the previous state. Further, the game does not punish students for making these errors. Namely, making such errors does not affect the number of clovers (rewards) students would get after solving the problem. Overall student’s game progress is demonstrated through the number of unlocked “worlds” on the math “tree”. Thus, FH2T makes all students, who went through all levels and gathered all knowledge, the winners.

FH2T has been found to improve learning and conceptual understanding of mathematical equivalence in middle school students (Chan et al., 2023; Decker-Woodrow et al., 2023). Moreover, playing a game-based version of the platform contributed to greater learning gain than playing a non-game-based version in elementary school students (Hulse et al., 2019). However, the FH2T effect on learners’ math anxiety still has to be examined.

A limited-feature version of ASSISTments

ASSISTments is a math tutoring platform where students receive tasks assigned by their teachers. Unlike FH2T which employs a more gamified design to engage students, ASSISTments asks students to solve more traditional math problems but provides instructional support such as immediate feedback and hints (Heffernan & Heffernan, 2014). For our analysis, we used data from a limited-feature version of ASSISTments, in which students could ask for hints during problem-solving (see Figure A4) and receive immediate feedback about their accuracy (see Figure A5). The teacher components provided by ASSISTments were not used in this study. The study used nine problem sets composed of problems aligned to similar math content introduced in FH2T that were selected and adapted from Engage NY (2014), Utah Math Project (2016), and Illustrative Math (2017).

Measures

Math Anxiety. Pre- and post-intervention levels of math anxiety were measured using a questionnaire of 13 items selected from the Math Anxiety Scale for Young

Children-Revised ($\alpha = .87$) ([Ganley & McGraw, 2016](#)), and administered to students on the first and the last days of intervention. Participants assessed how they usually feel in a hypothetical math-related scenario on a 4-point Likert-type scale (no = 0; not really = 1; kind of = 2; yes = 3). This questionnaire measured students' math anxiety through three subscales: negative reaction (4 items, e.g., "Math gives me a stomachache"), numerical in-confidence (3 reverse coded items, e.g., "I like doing math problems on the board in front of the class"), and worrying (6 items, e.g., "I get worried before I take a math test"). In our analysis, we used an average score of all three subscales as a measure of students' math anxiety.

Difference in math anxiety. Based on the personal pre- and post-intervention math anxiety outcomes, we calculated the difference between math anxiety levels before and after the intervention by subtracting the pre-intervention math anxiety value from the post-intervention math anxiety value for each participant.

Classroom average math anxiety. We calculated an average level of math anxiety in each participating classroom based on the personal pre-intervention math anxiety levels of all in-person students who participated in the larger study and completed the math anxiety questionnaire before the intervention (2088 students).

Pre-intervention math performance. Before the intervention, students completed an algebraic knowledge assessment consisting of 10 total problems. Each item was initially scored as correct or incorrect. For this variable, we used the sum of the correctly answered problems.

Classroom average math performance. We calculated an average level of math performance in each participating classroom based on the personal pre-intervention algebraic knowledge assessments of all in-person students who participated in the larger study and completed the algebraic knowledge assessment before the intervention (2088 students).

Gender. Gender information was provided by the school district.

Race/Ethnicity. Information regarding race/ethnicity was provided by the school district. All races were dummy coded separately and entered into the HLM models. As in the dataset only three races had more than 40 participants, only those three races were added into HLM models as three dummy coded variables: Hispanic/Latino (Hispanic), Asian (Asian), and White (White).

All variables used in the HLMs model can be found in Table 1.

Table 1 Variables Used in the Study

Measure	Variable name	Level	Coded
Pre-intervention math anxiety	PreMA	Student-level	Numeric
Post-intervention math anxiety	PostMA	Student-level	Numeric
Difference between pre- and post-intervention math anxiety	MADif	Student-level	Numeric
Type of the intervention: game-based (FH2T) or non-game-based (ASSISTments)	Gamify	Student-level	Dummy: 0 - ASSISTments, 1 - FH2T
Pre-intervention math performance	PrePerf	Student-level	Numeric
Gender	Male	Student-level	Dummy: 0 - Female, 1 - Male
Race, ethnicity: Hispanic	Hispanic	Student-level	Dummy: 0 - not Latino/Hispanic, 1 - Latino/Hispanic
Race, ethnicity: Asian	Asian	Student-level	Dummy: 0 - not White, 1 - White
Race, ethnicity: White	White	Student-level	Dummy: 0 - not White, 1 - White
Classroom average pre-intervention math anxiety	ClassMA	Classroom-level	Numeric
Classroom average pre-intervention math performance	ClassPrePerf	Classroom-level	Numeric

Procedure

Regardless of whether students participated in-person or virtually, all students individually completed their study assessments and gameplay sessions online using their own devices. Those students who attended in person were physically in the same classroom at the same time while participating in the study. Thus, in a given classroom, students completed their work either in FH2T, DragonBox, or ASSISTments.

Before their first gameplay session, students completed a 45-minute pre-intervention of their algebraic knowledge and math anxiety. Students then participated in nine 30-minute gameplay sessions on their assigned platform during their mathematics classes across the school year. After each session, students' progress was saved, so that they could continue the game/lesson in their subsequent

session. After their last gameplay session, students completed a post-intervention questionnaire measuring their math anxiety. The study was approved by an Institutional Review through exemption 1. Parental consent was not required but parents and guardians were given the opportunity to opt their child out of the study. Students assigned to the FH2T condition progressed through the game starting at World 1 in the same order. During each intervention session, they played FH2T for 30 minutes and solved as many problems as they could. After 30 minutes, the game logged them out automatically; their progress was saved so that they could start at the same problem in the next session. In the ASSISTments condition, students were assigned a problem set of 35 to 45 problems during each intervention session and worked through the problems in the same order. After 25 minutes or after students completed the assigned problem set, ASSISTments gave

students a session report that displayed their performance on each problem; students were instructed to review the report and revisit problems as needed.

Approach to Analysis

First, descriptive statistics and correlations were conducted for all the variables. For the first hypothesis, to detect the impact of both learning platforms on math anxiety we ran Welch's Two-Sample T-Tests between initial pre-intervention math anxiety and post-intervention math anxiety for each platform.

For the rest of the hypotheses, we employed lmerTest package in R to run 2-level Hierarchical Linear Models (HLM), with the first level being student variables, and the second level being the classroom variables. We specified the difference between pre- and post-intervention math anxiety as an outcome. We standardized math anxiety values using the pre-intervention grand mean pooled standard deviation. After running the null model and calculating the intraclass correlation coefficient which shows how much variance is explained by a random variation across the classrooms, we ran our first model with first-level variables that we aimed to control for, specifically gender, ethnicity, individual pre-intervention math performance, and individual pre-intervention math anxiety (we assumed that all of them had fixed effect across classrooms; see equation 1):

$$MADif_{ij} = (\gamma_{00} + u_{0j}) + \gamma_{10} * Gender_{ij} + \gamma_{20} * PrePerf_{ij} + \gamma_{30} * PreMA_{ij} + \gamma_{40} * Hispanic_{ij} + \gamma_{50} * Asian_{ij} + \gamma_{60} * White_{ij} + r_{ij} \quad (1)$$

where

γ_{00} – fixed effect of the intercept,

u_{0j} – random effects of the intercept in a specific group (j),

r_{ij} – error of specific variable (i) in a specific group (j).

Next, we added our main predictor – a dummy variable of the type of intervention: non-game-based platform (ASSISTments) or game-based platform (FH2T) – to the model. We assumed it had a random effect on the classroom level such that it affected the intercept and intervention coefficient (see equation 2).

$$MADif_{ij} = (\gamma_{00} + u_{0j}) + \gamma_{10} * Gender_{ij} + \gamma_{20} * PrePerf_{ij} + \gamma_{30} * PreMA_{ij} + \gamma_{40} * Hispanic_{ij} + \gamma_{50} * Asian_{ij} + \gamma_{60} * White_{ij} + (\gamma_{70} + u_{7j}) * Gamify_{ij} + r_{ij} \quad (2)$$

where

γ_{00} – fixed effect of the intercept,

u_{0j} – random effects of the intercept in a specific group (j),

u_{7j} – random effects of the type of the intervention in a specific group (j),

r_{ij} – error of specific variable in a specific group (j).

In the third model, we added classroom-level pre-intervention math anxiety and pre-intervention math performance as predictors to control for them, assuming they can only affect the intercept (see equation 3):

$$MADif_{ij} = (\gamma_{00} + u_{0j}) + \gamma_{10} * Gender_{ij} + \gamma_{20} * PrePerf_{ij} + \gamma_{30} * PreMA_{ij} + \gamma_{40} * Hispanic_{ij} + \gamma_{50} * Asian_{ij} + \gamma_{60} * White_{ij} + (\gamma_{70} + u_{7j}) * Gamify_{ij} + \gamma_{80} * ClassPreMA_j + \gamma_{90} * ClassPrePerf_j + r_{ij} \quad (3)$$

where

γ_{00} – fixed effect of the intercept,

u_{0j} – random effects of the intercept in a specific group (j),

u_{7j} – random effects of the type of the intervention in a specific group (j),

r_{ij} – error of specific variable in a specific group (j).

In the fourth model, we explored the effect of second level variable (average classroom pre-intervention math anxiety) on the model intercept and its contribution to the classroom-level variance of the intervention impact by adding multiplication of that variable and intervention type to the model (see equation 5):

$$MADif_{ij} = (\gamma_{00} + \gamma_{01} * ClassPreMA_j + u_{0j}) + \gamma_{10} * Gender_{ij} + \gamma_{20} * PrePerf_{ij} + \gamma_{30} * PreMA_{ij} + \gamma_{40} * Hispanic_{ij} + \gamma_{50} * Asian_{ij} + \gamma_{60} * White_{ij} + (\gamma_{70} + \gamma_{41} * ClassPreMA_j + u_{7j}) * Gamify_{ij} + \gamma_{80} * ClassPreMA_j + r_{ij} \quad (5)$$

where

γ_{00} – fixed effect of the intercept,

u_{0j} – random effects of the intercept in a specific group (j),

u_{7j} – random effects of the type of the intervention in a specific group (j),

r_{ij} – error of specific variable in a specific group (j).

Results

Before running the data analysis, we calculated means and standard deviations for all math anxiety measures and the math performance measure in different study conditions (see Table 2) and prepared a correlation matrix for all quantitative variables from our models (see Table 3). Our data demonstrated a negative correlation between math anxiety and performance (see Table 3), which is in accordance with other studies (Barroso et al., 2021).

Table 2 Means (and Standard Deviations) of Math Anxiety and Performance Measures in Different Conditions

Groups	N	Pre-intervention performance	Pre-intervention math anxiety	Post-intervention math anxiety	Math anxiety difference ^a	Average classroom pre-intervention math anxiety
ASSISTments	222	4.02 (2.18)	14.61 (5.48)	14.23 (5.89)	-.38 (4.97)	14.14 (1.99)
FH2T	455	3.92 (2.26)	14.30 (5.89)	14.21 (6.23)	-.09 (4.46)	14.16 (2.07)
Total	677	3.95 (2.23)	14.40 (5.76)	14.22 (6.16)	-.18 (4.63)	14.15 (2.04)

^a Math anxiety difference is the difference between pre-intervention and post-intervention math anxiety calculated by subtracting pre-intervention values from the post-intervention values.

Table 3 Means, Standard Deviations, and Correlations with Confidence Intervals (before standardization)

Variable	<i>M</i> ^a	<i>SD</i>	1	2	3	4	5
1. Pre-intervention performance	3.95	2.23					
2. Pre-intervention math anxiety	14.40	5.76	-.12** [-.19, -.04]				
3. Post-intervention math anxiety	14.22	6.12	-.09* [-.17, -.02]	.70** [.66, .73]			
4. Difference between pre- and post-intervention math anxiety	-0.18	4.63	.02 [-.05, .10]	-.32** [-.39, -.25]	.45** [.39, .51]		
5. Average classroom pre-intervention math anxiety	14.15	2.04	-.15** [-.22, -.07]	.36** [.29, .42]	.23** [.16, .30]	-.14** [-.21, -.07]	
6. Average classroom pre-intervention math performance	3.85	1.55	.72** [.68, .75]	-.09* [-.16, -.01]	-.04 [-.12, .04]	.05 [-.02, .13]	-.25** [-.32, -.18]

^a *M* and *SD* are used to represent mean and standard deviation, respectively. Values in square brackets indicate the 95% confidence interval for each correlation.

* indicates $p < .05$. ** indicates $p < .01$.

RQ1. How does participating in game-based and non-game-based digital learning interventions affect students' math anxiety?

Our first hypothesis for this research question was that math anxiety would decrease after both – game-based and non-game-based interventions (i.e. FH2T and

ASSISTments). This hypothesis was not supported: t-tests between pre- and post-intervention of math anxiety levels demonstrated no statistically significant change in math anxiety in FH2T $t(454) = .03, p = .98$ nor in ASSISTments $t(221) = .90, p = .37$.

For the second hypothesis, we believed that math anxiety would decrease more after FH2T than after ASSISTments. In contrast to our hypothesis, FH2T did

not demonstrate a larger effect on personal math anxiety than ASSISTments: $t(677) = 0.7, p = .48$ (see Table 4).

RQ2. Do classroom characteristics impact the effect of the intervention type (game-based or not game-based) on post-intervention math anxiety?

Our first hypothesis for this research question was that the effect of the intervention type would largely vary across classrooms. The effect of the intervention condition did vary between classrooms with a standard deviation of the effect SD (d) = 0.21 between classrooms (see Table 4 and Figure B1), and an effect size of 0.2 or greater is considered large in educational research in accordance with Kraft's benchmarks (Kraft, 2020). For the second hypothesis, we believed that classroom average pre-intervention math anxiety would moderate the association between the intervention type and individual post-intervention anxiety, so that students in

higher math-anxious classrooms would demonstrate a steeper decrease in math anxiety in a game-based condition compared to a more classical math learning condition than students in a less math anxious classrooms. Results showed that classroom average pre-intervention math anxiety significantly moderated the association between intervention type and the decrease in personal math anxiety $t(677) = 2.5, p = .014$. However, the directionality of that change was contrary to what we expected. Namely, the effect size of intervention type on post-intervention math anxiety increased by 0.18 Cohen's d for every one standard deviation increase in classroom average math anxiety. While this is a small effect, it is notable that it is statistically significant. Specifically, ASSISTments tended to decrease personal levels of math anxiety more in classrooms with higher levels of average math anxiety. On the other hand, FH2T did not demonstrate such a tendency and affected classrooms comparably with different math anxiety levels (see Figure 1).

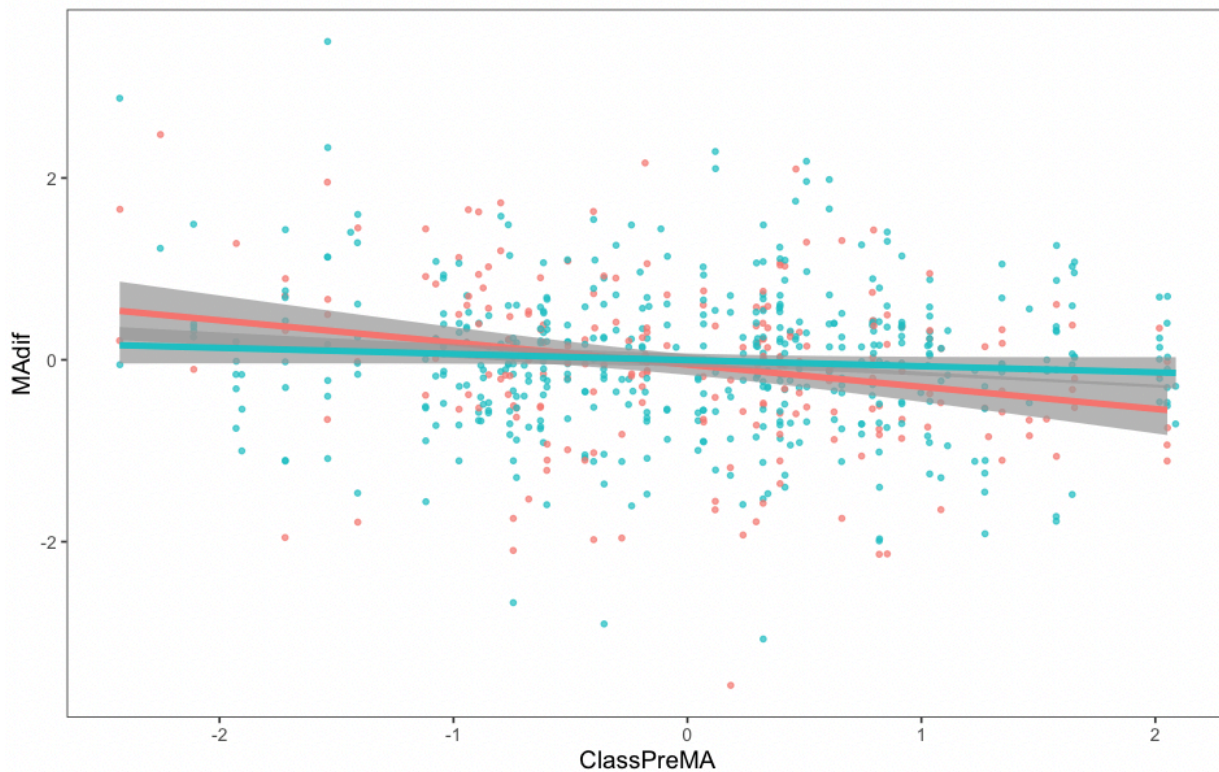


Figure 1 Moderation Effect of Game-Based Condition on Association between Difference in Math Anxiety and Classroom Average Math Anxiety.

The red line represents results for ASSISTments, the blue line represents results for FH2T. Values were standardized using the pre-intervention grand mean pooled standard deviation.

Table 4 Hierarchical Linear Models Predicting Difference Between Pre- and Post-Intervention Levels of Math Anxiety

Predictors	Null Model Estimate (CI)	Model 1 Estimate (CI)	Model 2 Estimate (CI)	Model 3 Estimate (CI)	Model 4 Estimate (CI)
Intercept	- 0.01 (-0.09 – 0.07)	0.06 (-0.18 – 0.31)	- 0.01 (-0.26 – 0.25)	- 0.05 (-0.35 – 0.25)	- 0.07 (-0.36 – 0.21)
Male		- 0.15 (-0.28 – -0.03) *	- 0.17 (-0.29 – -0.04) **	- 0.16 (-0.28 – -0.04) *	- 0.17 (-0.30 – -0.05) **
PrePerf ^a		0.01 (-0.04 – 0.02)	- 0.01 (-0.04 – 0.02)	- 0.03 (-0.07 – 0.201)	- 0.02 (-0.06 – 0.01)
PreMA ^b		- 0.28 (-0.34 – -0.22) ***	- 0.29 (-0.35 – -0.23) ***	- 0.28 (-0.34 – -0.22) ***	- 0.28 (-0.35 – -0.22) ***
Asian		0.09 (-0.24 – 0.41)	0.09 (-0.23 – 0.42)	0.07 (-0.25 – 0.40)	0.06 (-0.26 – 0.39)
Hispanic		0.07 (-0.19 – 0.32)	0.08 (-0.18 – 0.33)	0.07 (-0.19 – 0.32)	0.09 (-0.16 – 0.35)
White		0.09 (-0.13 – 0.32)	0.10 (-0.12 – 0.33)	0.10 (-0.13 – 0.32)	0.11 (-0.12 – 0.33)
Gamify			0.05 (-0.08 – 0.18)	0.06 (-0.07 – 0.19)	0.05 (-0.08 – 0.18)
Class PreMA ^c				- 0.03 (-0.11 – 0.05)	- 0.14 (-0.25 – -0.03) *
Class PrePerf				0.03 (-0.03 – 0.10)	0.03 (-0.02 – 0.09)
Class PreMA * Gamify					0.18 (0.04 – 0.32) *
N	677	677	677	677	677
Random effects					
σ^2	0.63	0.57	0.58	0.56	0.57
τ_{00}	0.23	0.21	0.00	0.21	0.00
τ_{11}			0.21	0.17	0.21
ICC	0.08				
N of students	677	677	677	677	677
N of classrooms	98	98	98	98	98
Marginal R ²	0.00	0.11	0.12	0.12	0.13

* – p-value < .05, ** – p-value < .01, *** – p-value < .001.

^a PrePerf refers to Pre-intervention performance.

^b PreMA refers to Pre-intervention math anxiety, math anxiety values are standardized.

^c Class refers to Class average values.

Discussion

Our study examined the effect of game-based (From Here To There) and non-game-based (ASSISTments) math learning interventions on students' math anxiety. Contrary to our hypotheses, results did not show a significant impact of either intervention on decreasing students' math anxiety. We did find, however, that the impact of the interventions on math anxiety varied significantly across classrooms. Below, we discuss how the study context might have impacted these results.

Student Level Math Anxiety Did Not Change After Either Intervention

Our first hypothesis was not supported, namely, students' levels of math anxiety did not significantly change after either intervention. Further, in contrast to our second hypothesis, no significant difference was found between game-based – From Here To There (FH2T) – and non-game-based – ASSISTments – interventions' effects on post-intervention math anxiety. On the one hand, these results are in accordance with other studies that did not report positive effects of digital math learning interventions against math anxiety (e.g., see the report by Atoyebi & Atoyebi (2022)) and, specifically, game-based math learning interventions (e.g., Hung et al., 2014; Jansen et al., 2013). Thus it might be that these types of interventions are not effective tools for decreasing math anxiety. Indeed, the majority of studied math learning platforms, including FH2T, were created not to specifically target math anxiety but rather to focus on boosting math learning.

Another potential explanation is that there was not enough dosage to impact students' overall levels of generalized math anxiety. The study was only conducted during nine 30-minute sessions throughout the school year. Researchers distinguish two types of math anxiety: trait-level math anxiety which refers to a stable negative predisposition towards math-related situations as threatening (Daches Cohen et al., 2021), and state math anxiety which refers to an in-the-moment physiological reaction to math activities (Strohmaier et al., 2020). Self-reported measures of math anxiety – the type of math anxiety measure that we used – were reported to be significantly associated with trait-level math anxiety (Conlon et al., 2021; Strohmaier et al., 2020).

However, given that the intervention only consisted of nine 30-minute sessions, it could be that trait math anxiety needs more prolonged or intense intervention to be impacted. On the other hand, the intervention could have affected state math anxiety which in case of a long-term exposure, can affect trait math anxiety, as well. Thus, even if while working on FH2T or ASSISTments students experienced less nervousness than during more traditional math classes, this state math

anxiety difference could be not detected by the math anxiety measure used in the study, and intervention duration was not enough to impact trait math anxiety.

On the other hand, it is noteworthy to mention the context in which our study was conducted. The study was implemented in the 2020 – 2021 school year, which was the height of the COVID-19 pandemic. Much research from the pandemic showed that math performance dropped significantly during this time (Kuhfeld et al., 2022) and students across the globe reported a severe increase in their generalized anxiety during that period (Hawes et al., 2021). As math anxiety has been associated with both math performance (Barroso et al., 2021) and general anxiety (Szczygie, 2020), we might expect that math anxiety should have naturally increased during this time. However, we did not find significant changes in students' math anxiety levels during this time after either intervention. Additional research is needed to tease apart how these interventions may affect math anxiety in a non-COVID context, allowing greater speculation of our current results.

The Game-Based Platform Effect Varied Across Classrooms

To analyze the differences in intervention type effect on math anxiety more precisely, we further investigated the variation of this difference on the classroom level as well as the potential moderation effect of average classroom-level math anxiety. Variation of the intervention effect across classrooms could be considered large; thus characteristics of participants' classrooms did largely affect the difference between the two interventions' effectiveness against math anxiety.

Furthermore, a small but statistically significant cross-level interaction between classroom average math anxiety and intervention type was found. Specifically, in the classrooms with a lower classroom average level of math anxiety, neither intervention demonstrated any difference in math anxiety levels in students. This could be because most students in those classrooms are already comfortable with engaging with mathematical problems, thus they had less math anxiety to reduce. On the other hand, students whose classmates have a higher average level of math anxiety demonstrated a slightly larger decrease in math anxiety after the non-game-based intervention (ASSISTments) than after the game-based intervention (FH2T), contradicting our hypothesis. There are a few potential explanations for such an effect: firstly, it could be that students in these classrooms could have less prior experience with non-traditional math learning activities and facing such a novel type of math activity could have increased their math anxiety. Secondly, classrooms with higher levels of math anxiety may have a distinct peer climate and students' reactions to intervention could have been influenced by the more

math-anxious reactions demonstrated by their peers. Those explanations are speculative but might inform future studies, thus we describe them in detail in the “Implications and Future Directions” section below.

Limitations

An important limitation of our study is the fact that intervention was conducted during the first years of COVID-19, providing students with an extra stressful context. As these stressful events might affect anxiety levels of students in general and academic-related anxieties in particular, it is hard to distinguish the impact of our interventions from the pandemic context as a whole and make conclusions regarding FH2T and ASSISTments’ effectiveness against math anxiety in students.

Another limitation relates to the math anxiety measure that was utilized and the longevity of the intervention. As discussed above, the self-reported math anxiety measure might have not revealed whether interventions affected state math anxiety they experienced while playing the math learning game. Furthermore, more prolonged intervention might be needed to affect a trait math anxiety – a stable attitude towards math.

Finally, our study examined only two math learning platforms – one for game-based condition (FH2T) and one for non-game-based condition (ASSISTments) – which prohibits us from making generalizations about the difference between all game-based and non-game-based learning platforms. Future research would add meaningful contributions by following up on this work with different systems to identify further nuances that may contribute to changes in math anxiety.

Implications and Future Directions

One future direction that can build on the current work is to assess which classrooms would benefit from a specific type of digital intervention compared to other interventions. Based on our results, classroom characteristics are important and need to be considered as the effect of condition type (game-based vs non-game-based) on math anxiety largely varies across classrooms. Specifically, class-average math anxiety showed a small but significant moderation effect on the

association between intervention type and students’ change in math anxiety.

One potential explanation might be that game-based platforms present a context that is *too* novel for math-anxious students. Some past research has found that math-anxious students often stick to familiarity, even when it is not efficacious to do so (Ramirez et al., 2016), indicating that a break away from familiar contexts might make students more anxious. At the classroom level, teachers might be inclined to provide their students with more traditional math learning activities to combat the anxiety of the class. In this case, when facing unusual game-based interventions during our research, students who had little prior experience with these kinds of interventions may have experienced greater feelings of anxiousness due to the need to adjust to the new type of activity, worsening math anxiety.

Moreover, peer influence in the classroom might also explain this finding (but in the opposite direction to what we initially suggested). For example, students in classrooms with higher average classroom level math anxiety may be surrounded by students who have a more anxious perception towards game-based math platforms. Thus, they could be prone to becoming more anxious after working with this type of platform, even if their personal pre-intervention level of math anxiety was not very high. However, this would need to be explicitly tested. Moreover, future work in this area would add meaningful contributions by further measuring both trait and state math anxiety, as it would provide greater nuance to shed light on complicated relationships among game-based math learning interventions and math anxiety.

Conclusion

This study investigated the effect of game-based and non-game-based digital math learning platforms on math anxiety and its variation across classrooms. Results showed no changes in math anxiety in either game-based or non-game-based intervention conditions but found that these effects can vary between classrooms. It is worth focusing future research on classroom-level differences that might impact game-based and non-game-based digital intervention effects on students’ math anxiety.

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Appendix A

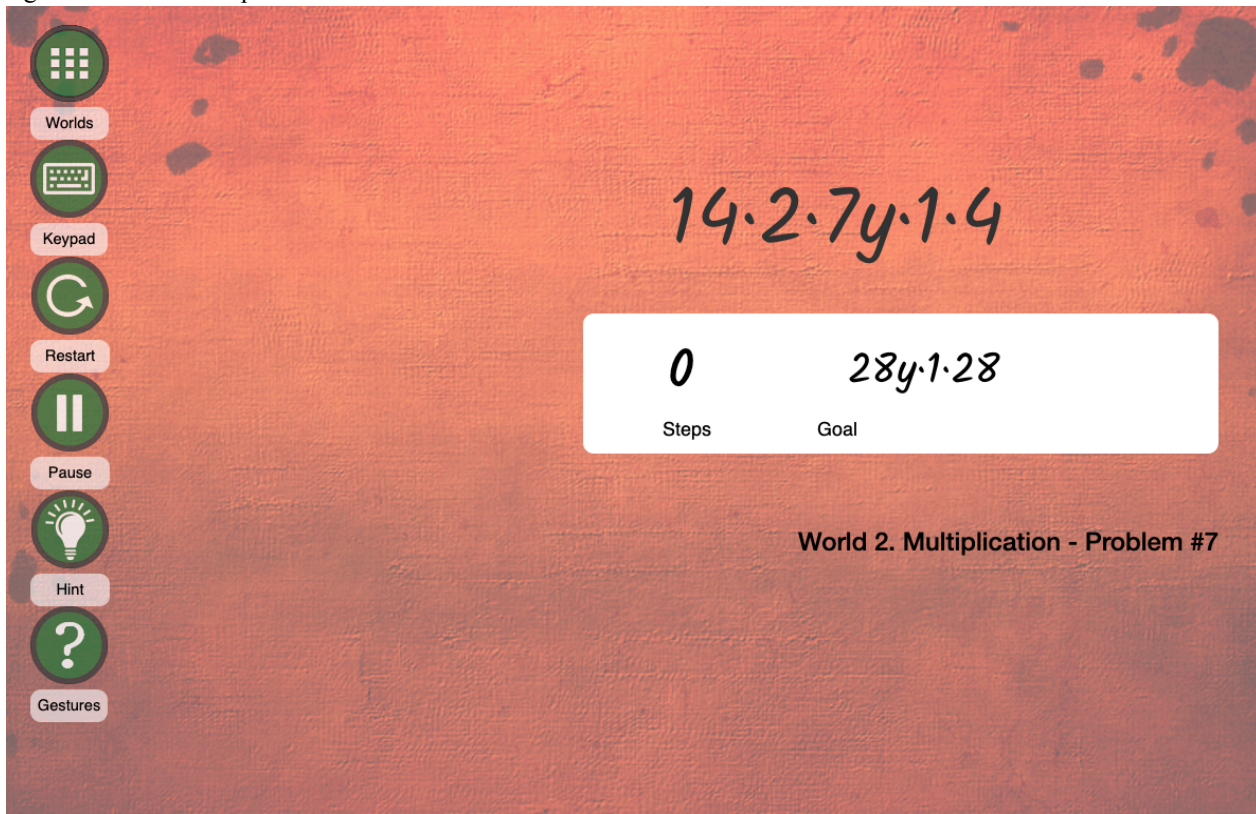
FH2T and ASSISTments Screenshots

Figure 1. FH2T: the main page – overview of all the math worlds



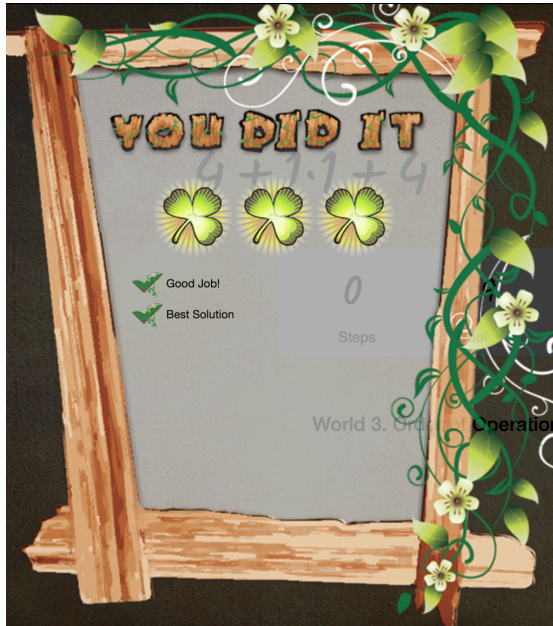
Note. This screen demonstrates the state when the user has already achieved three math “Worlds”.

Figure 2. FH2T: math problem screen

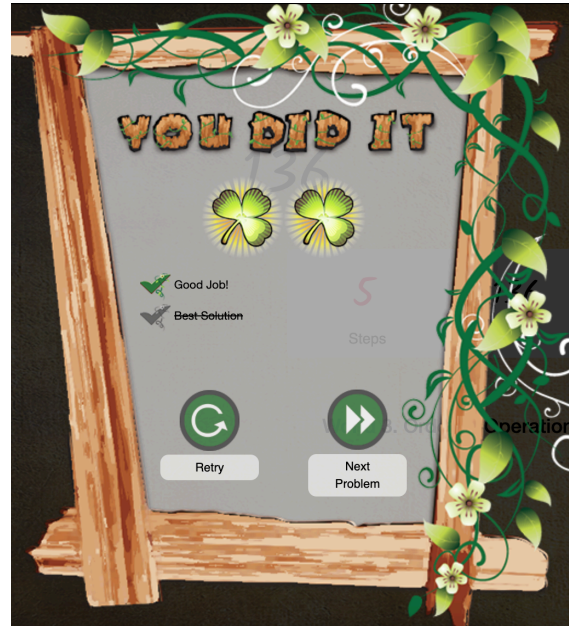


Note. On this screen, the user needs to transform “ $14 \times 2 \times 7y \times 1 \times 4$ ” to the goal state “ $28y \times 1 \times 8$ ” by dragging numbers with the computer mouse to appropriate places.

Figure 3. FH2T: feedback after each problem solving



1. Problem solved in the most effective way



2. Problem solved not in the most effective way

Figure 4. ASSISTments: hints in each problem

The screenshot displays the ASSISTments web application interface. At the top, a yellow header bar contains the 'ASSISTments' logo and navigation links for 'Teacher', 'Student', and 'Builder'. On the right of the header, there is an 'Account' section for 'Dr. Jhake (jhake@wpi.edu)' with a 'Logout' link. Below the header, a dark blue bar contains 'Settings' and 'About' links. The left sidebar shows 'Problems completed: 0/1' and a list of problems, with 'Calculate the di...' being the selected one. The main content area is titled 'Assignment: Problem #PSABKETV'. It displays the problem ID 'PRABKETV' and the instruction 'Calculate the difference. $-2 - (-16)$ '. A yellow hint box contains the text 'Subtracting a negative number is the same as adding a positive number.' with a 'Comment on this hint' link. A red arrow labeled 'Example Hint' points to this box. Below the hint, there is a text input field and a 'Submit Answer' button. At the bottom right, a progress bar shows '75%' completion. A red arrow labeled 'Request Hints' points to a 'Show hint 2 of 3' button.

ASSISTments Teacher Student Builder Account Dr. Jhake (jhake@wpi.edu) Logout

Settings About

Problems completed: 0/1

Calculate the di...

Assignment: Problem #PSABKETV

Problem ID: PRABKETV

Calculate the difference.
 $-2 - (-16)$

engage™

Subtracting a negative number is the same as adding a positive number.

Comment on this hint

Type your answer below as a number (example: 5, 3, 1, 4 1/2, or 3/2):

Submit Answer

75% ?

Request Hints Show hint 2 of 3

Calculate the difference.
 $-2 - (-16)$
[Comment on this problem](#)

engageTM

Subtracting a negative number is the same as adding a positive number.
[Comment on this hint](#)

An example:
 $-2 - (-1)$
 $= -2 + 1$
 $= -1$
[Comment on this hint](#)

$-2 - (-16)$
 $= -2 + 16$
 $= 14$
[Comment on this hint](#)

Correct answer:
14

← **Correct Answer**

Type your answer below as a number (example: 5, 3.1, 4 1/2, or 3/2):

Submit Answer

Show answer →

Show answer

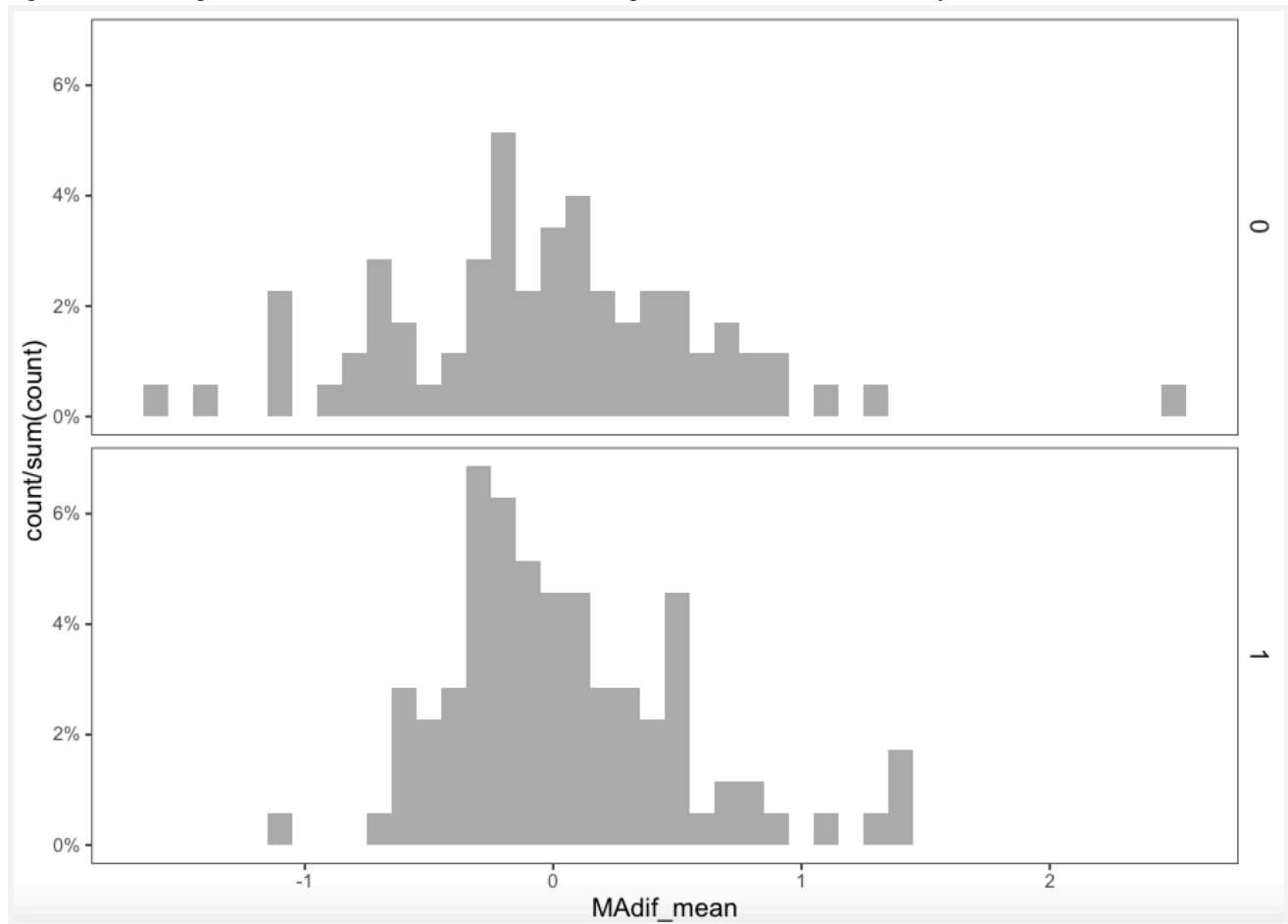
0% [?]

22

Appendix B

Variation of Pre- to Post-Intervention Math Anxiety Difference in Game-Based vs Non-Game-Based Conditions

Figure 1. Percentages of classrooms demonstrated different post-intervention math anxiety levels



Note. Upper histogram demonstrates variation in post-level math anxiety in ASSISTments condition, lower histogram – variation in post-level math anxiety in FH2T condition. Y-axes shows the percentage of classes, X-axes – level of pre- to post-intervention math anxiety difference.

Math anxiety values were standardized using the pre-intervention grand mean pooled standard deviation.